

GlassBox

Agentic, but auditable.

An autonomous options agent whose every order passes a deterministic governor, with a ledger that replays any decision.

One function is the seam.



`build_candidates` is the one-function seam where the MCP strategist plugs in. Nothing downstream cares who proposed: the governor recomputes every number it is handed.

DEFINED-RISK ONLY

The order builder cannot express a naked short.

The covering asset is a required argument with no default. Covered calls, cash-secured puts, defined-risk verticals — and iron condors exist only as two governed verticals.

The governor recomputes max loss from strikes, ratio, quantity and net price, and reads the strategist's claim only to record the divergence.

Paper in the URL on every request.

DATA

Free paper tier for the full SPY chain: contracts paginated to exhaustion, snapshots on the indicative feed, clock and calendar for as_of.

MCP

Alpaca's MCP server sits in the natural-language strategist path, where a model reasoning about the market belongs.

EXECUTION

Official alpaca-py SDK with an account identity guard; idempotent client order ids embed the root ledger id.

GUARD

The executor reads the account number over the connection that would carry the order and refuses if it is not the one this run was authorised for.

THE NUMBERS

Small, governed, replayable.

189

contract tests across 7 suites —
the merge gate

5 lots

of a governed 763/758 SPY put
vertical, every leg expiring inside
the scored window

2,067

total computed max loss on a
10,000 book cap

And one refusal on the ledger, replayed `matched = True`.

TEN CHECKS, EVERY DECISION

Seven pinned by the seam, three extensions.

structure_valid

max_loss_cap

2% of equity per structure

cash_floor

20% of equity

market_open

x_max_expiry

on or before 2026-09-03

net_reconciles

coverage

churn_guard

3,600 s window · 7,200 s min hold

x_position_cap

4 open · 2 per underlying

x_total_open_risk

10% of equity across the book

The agent refused its own trade.

CLAIMED MAX LOSS

250.00

the strategist's figure

COMPUTED MAX LOSS

152,584.00

the governor's own arithmetic

CAP

2,000.00

2% of equity

DIVERGENCE

152,334.00

written into the record

**GOVERNOR
REJECTED**

Refused on `max_loss_cap · coverage · cash_floor ·
x_total_open_risk`. Written down. Replayed **matched = True**.

TWO PODS, ONE SEAM

Japan and the United States, coordinated only through the repo.

Two humans, each with an AI builder pod, on opposite schedules. The pods never talk to each other. GB_INTERFACES.md is signed by both humans; contract tests are the neutral arbiter; the baton is a written HANDOFF.

Pre-event work disclosed in the README. MIT licensed. Paper trading only.